

BEISEKER AGCM, AB, Canada

WMO: 710090

Lat: 51.379N

Lon: 113.357W

Elev: 2940

StdP: 13.20

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-23.7	-17.1	-29.3	1.2	-23.6	-22.6	1.8	-16.8	27.6	22.6	25.0	18.4	4.0	350	0.688

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.3	84.5	61.9	81.0	60.4	77.7	59.6	66.0	77.9	63.9	75.7	61.8	73.6	8.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.6	91.5	72.7	59.3	84.0	69.8	57.1	77.5	66.9	32.4	77.6	30.7	75.7	29.1	73.6	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.7	21.9	18.8	-32.0	91.4	6.5	3.8	-36.7	94.2	-40.5	96.4	-44.2	98.5	-48.9	101.3
			-32.2	72.3	6.5	2.8	-36.9	74.3	-40.7	76.0	-44.3	77.6	-49.0	79.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.6	15.5	13.8	25.3	37.5	49.8	56.8	60.8	59.7	51.5	39.3	25.5	13.8
	DBStd	20.42	14.91	14.72	14.02	9.01	7.57	5.29	4.46	5.34	8.13	8.78	12.45	14.49
	HDD50	5618	1070	1014	766	381	98	8	0	3	76	343	736	1123
	HDD65	10042	1535	1434	1231	824	473	250	143	177	406	796	1185	1588
	CDD50	1085	0	0	0	6	91	213	335	304	122	13	1	0
	CDD65	35	0	0	0	0	1	5	13	13	3	0	0	0
	CDH74	1642	0	0	0	11	126	180	490	547	278	10	0	0
	CDH80	399	0	0	0	1	14	31	118	158	76	1	0	0
	WSAvg	7.4	7.5	7.1	7.8	8.8	9.1	8.4	6.5	6.1	6.8	7.2	7.0	6.6
	PrecAvg	20.3	0.7	0.6	1.0	1.5	2.7	3.9	2.9	2.5	1.8	1.0	0.9	0.8
(15) Precipitation	PrecMax	26.7	1.5	1.3	2.1	3.4	4.3	10.0	6.3	4.5	4.3	2.0	2.4	1.7
	PrecMin	12.9	0.2	0.1	0.3	0.6	0.9	1.7	1.1	0.5	0.3	0.3	0.2	0.2
	PrecStd	2.7	0.3	0.3	0.4	0.7	1.0	1.5	1.2	0.9	0.9	0.4	0.5	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.3	45.7	60.6	73.5	81.9	83.6	87.6	89.7	86.8	74.1	60.9	47.4
		MCWB	38.5	38.4	45.5	50.0	55.0	61.1	66.1	61.8	58.5	52.6	45.8	36.9
	2%	DB	41.6	41.0	54.0	66.4	77.8	78.8	83.6	84.1	81.7	66.4	52.6	41.8
		MCWB	35.9	35.2	41.3	47.6	53.9	59.1	64.6	62.2	57.3	49.9	41.9	35.5
	5%	DB	39.3	37.5	48.6	60.3	73.0	75.1	79.3	80.1	76.4	61.5	46.7	37.9
		MCWB	34.6	33.1	38.7	43.9	52.1	57.7	63.6	61.2	55.8	47.0	37.9	32.9
	10%	DB	36.2	33.8	42.7	54.9	68.3	71.8	75.7	75.9	70.8	56.2	41.9	33.9
		MCWB	32.4	30.6	36.2	41.5	50.4	56.0	62.1	60.3	54.0	44.6	35.5	30.1
	0.4%	WB	38.4	39.2	46.5	51.6	56.6	65.5	70.9	67.3	60.8	54.2	46.9	39.3
		MCDB	44.7	45.1	58.8	71.1	76.8	77.4	81.3	78.5	80.4	70.9	59.9	44.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.5	35.8	42.5	47.9	54.7	62.0	67.3	65.0	58.7	50.8	42.3	36.1
		MCDB	41.5	40.4	52.0	65.1	73.6	74.6	78.6	77.8	76.7	64.5	51.4	41.2
	5%	WB	34.8	33.3	39.1	44.8	53.1	59.5	64.9	63.1	56.5	47.9	39.0	32.9
		MCDB	39.2	36.9	47.4	57.9	70.2	71.0	76.3	75.5	73.7	59.7	45.8	37.4
	10%	WB	32.3	30.9	36.3	42.2	51.3	57.4	62.9	61.0	54.5	45.2	36.0	30.1
		MCDB	36.2	33.8	42.2	53.0	65.5	68.7	73.8	73.6	69.9	55.3	41.0	34.1
(35) Mean Daily Temperature Range	5% DB	MDBR	20.0	20.5	21.2	24.5	26.5	24.5	27.3	28.8	29.4	26.7	21.8	20.0
		MCDBR	20.4	23.0	28.8	36.4	36.5	32.1	35.0	38.1	41.4	35.3	28.1	22.3
		MCWBR	15.4	17.9	17.1	19.2	16.1	15.5	19.6	18.4	19.1	19.6	17.6	16.8
	5% WB	MCDBR	19.8	21.7	27.3	33.8	31.8	28.1	32.0	32.8	38.0	33.5	27.0	22.0
		MCWBR	15.1	17.1	17.2	18.2	14.3	15.1	19.1	17.8	18.0	19.0	17.8	16.9
(40) Clear-Sky Solar Irradiance	taub		0.258	0.270	0.281	0.332	0.349	0.346	0.346	0.362	0.317	0.281	0.252	0.238
	taud		2.374	2.423	2.471	2.342	2.352	2.401	2.411	2.365	2.451	2.534	2.487	2.383
	Ebn at Noon		248	278	295	288	287	288	286	275	278	271	252	239
	Edh at Noon		19	24	28	36	37	36	35	35	28	21	16	16
(44) All-Sky Solar Radiation	RadAvg		353	677	1111	1507	1822	1957	2094	1692	1192	722	386	276
	RadStd		26	56	94	117	121	118	103	119	123	84	32	27

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (71 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page